

David Matos

Software / Backend Engineer

Porto, Portugal | +351 917337707 | davidmatos0417@gmail.com | linkedin.com/in/davidmatos | github.com/davidmatos17 | Portfolio

SUMMARY

Computer Science graduate with hands-on experience in **Python backend development**, **data processing**, **automated testing**, **CI/CD**, and **cloud-native technologies**. Developed reusable software adopted by internal engineering teams, combining a **problem-solving mindset** with strong **teamwork**, **adaptability**, and **discipline** developed through 10 years of competitive team sports.

EDUCATION

Faculty of Sciences, University of Porto

Porto, Portugal

B.Sc. in Computer Science

2022 – 2026

- Relevant coursework: Software Engineering, Distributed Systems, Algorithms, Databases, Large-Scale Data Science, Machine Learning, and Problem Solving.

EXPERIENCE

Curricular Intern – Software Engineering

October 2025 – January 2026

Celfinet, a Cyient Company

Lisbon, Portugal

- Designed and developed a reusable **Python library with hundreds of functions** to simplify and accelerate access to frequently used datasets for ad-hoc studies.
- Built reusable database-access abstractions using **SQLAlchemy**, reducing the complexity of recurring data-access workflows and improving consistency across studies.
- Implemented **automated testing and CI/CD** for testing, packaging, and internal distribution, enabling other developers to easily install and use the library.
- Worked closely with internal developers to refine the library around their recurring data-access needs; the solution was adopted by **two internal teams** and remained in use after the internship.

Summer Intern – Software Engineering

Summer 2025

Celfinet, a Cyient Company

Lisbon, Portugal

- Refactored and restructured an existing **FastAPI backend** for base-station energy consumption studies, improving code organization, maintainability, and component reusability.
- Developed and maintained reusable **Python components across dozens of internal packages** for processing very large datasets, improving code reuse and maintainability across data-intensive workflows.
- Designed and implemented a **CSV data validation and error-reporting system** to identify data-quality issues, including missing values, invalid data, incorrect units, and duplicate records, with automated correction where reliable rules could be applied.
- Worked across backend and data-processing tasks, maintaining **automated tests** and reusable components while supporting the adoption of developed packages by an internal team.

Vice President

2023 – 2024

NUCC – Student Nucleus of Computer Science

Faculty of Sciences, University of Porto

- Organized technical and career-oriented Job Fairs connecting Computer Science students with industry professionals and external organizations.
- Coordinated communication and execution across **students, faculty members, industry professionals**, and **external organizations**.
- Managed multiple initiatives within the Computer Science student community, developing practical experience in coordination, communication, and leadership.

PROJECTS

Personal Cloud Platform for Large-Scale Data Science | *Python, Docker, Kubernetes, Linux* 2026

- Designed and implemented a cloud-based platform for large-scale data science workloads using containerized services and cloud-native technologies.
- Used **Docker** and **Kubernetes** to package, deploy, and orchestrate application components, exploring container orchestration and scalable system deployment.
- Designed the system with a focus on **modularity, scalability, and automated deployment** within a Linux-based environment.

K-Nearest Neighbors (KNN) Model Optimization | *Python, Machine Learning, scikit-learn* 2026

- Implemented and evaluated the **K-Nearest Neighbors (KNN)** algorithm in Python, analyzing model performance across different datasets and configurations.
- Applied **K-Fold cross-validation** to evaluate model generalization and compare different KNN configurations using consistent evaluation procedures.
- Investigated the impact of **feature scaling, PCA dimensionality reduction, and Manhattan distance** on model performance and classification results.
- Compared different preprocessing and model configurations, analyzing their impact on **accuracy, generalization, and computational performance**.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript
Backend: FastAPI, Flask, Django, REST APIs, SQLAlchemy
Cloud & DevOps: Docker, Kubernetes, CI/CD, Linux, Bash, Azure, GCP
Software Engineering: Problem Solving, Object-Oriented Programming, Data Structures & Algorithms, Automated Testing, Git, Version Control
AI & Productivity: Prompt Engineering, AI-Assisted Development, LLM-based Tools
Data & Machine Learning: Data Processing, Data Analysis, Feature Engineering, Machine Learning Fundamentals, Model Evaluation
Tools: Postman
Languages: Portuguese (Native), English (Professional), Spanish (Conversational)

ADDITIONAL EXPERIENCE

High Performance Handball Player 2012 – 2022
Colégio Internato dos Carvalhos (GDCIC) Portugal

- 10-year competitive career including two national championships; built strong teamwork, discipline, and performance-under-pressure skills applicable to high-intensity engineering environments.